Comprehensive Study Guide: Vaccination & Control Strategies

Course: STA 6222 – Models for Disease Transmission & Management **Week:** 6

1. Introduction & Context

This week bridges the gap between **descriptive modeling** (describing how a disease spreads) and **intervention modeling** (describing how we stop it).

Previously, we treated parameters like β (transmission rate) and γ (recovery rate) as fixed biological constants. In this module, we view them as targets for control:

- **Vaccination** acts on the Susceptible pool (S) or the effective transmission potential (R_e).
- **NPIs (Non-Pharmaceutical Interventions)** act on the transmission rate (β).
- **Treatment** acts on the recovery rate (γ).

2. Deterministic Control Theory

2.1 The Concept of Herd Immunity

Herd immunity occurs when a sufficient proportion of a population is immune to a disease, making its spread from person to person unlikely. This protects individuals who are not immune (e.g., newborns, immunocompromised).

The Logic:

1. The epidemic threshold is $R_0 > 1$.
2. Vaccination reduces the effective reproduction number to R_e.
3. Control is achieved when we force $R_e < 1$.

2.2 Deriving Critical Coverage (p_c)

This is a standard exam derivation. You should be able to reproduce it.

Scenario A: Perfect Vaccine (100% Efficacy)

- Let p be the proportion vaccinated.
- The proportion remaining susceptible is $(1 - p)$.
- The effective reproduction number becomes:

$$R_e = (1 - p)R_0$$

- Set the control condition $R_e < 1$:

$$(1 - p)R_0 < 1$$

- Rearrange for p:

$$1 - p < \frac{1}{R_0} \implies p > 1 - \frac{1}{R_0}$$

- **Result:** $p_c = 1 - \frac{1}{R_0}$

Scenario B: Imperfect Vaccine (Efficacy $e < 1$)

- Vaccinating a proportion p only protects $p \times e$ people.
- The susceptible fraction is now $1 - pe$.
- The effective reproduction number is:

$$R_e = (1 - pe)R_0$$

- Set $R_e < 1$:

$$(1 - pe)R_0 < 1$$

$$1 - pe < \frac{1}{R_0}$$

$$pe > 1 - \frac{1}{R_0}$$

- **Result:** $p_c = \frac{1-1/R_0}{e}$

Critical Insight: If the vaccine efficacy e is *lower* than the required herd immunity threshold ($1 - 1/R_0$), the numerator becomes larger than the denominator ($p_c > 1$). This means **herd immunity is mathematically impossible**, even if you vaccinate 100% of the population.

3. The SIRV Model (Routine Vaccination)

While the algebra above applies to a "one-shot" campaign, real public health relies on routine vaccination of newborns. We use the **SIRV** differential equations.

3.1 The Equations

$$\frac{dS}{dt} = \mu N - \beta SI - \nu S - \mu S$$
$$\frac{dV}{dt} = \nu S - \mu V$$
$$\frac{dI}{dt} = \beta SI - \gamma I - \mu I$$
$$\frac{dR}{dt} = \gamma I - \mu R$$

Key Terms:

- μN: New births entering the system (assuming constant population N).
- $-\nu S$: Susceptibles leaving the S compartment due to vaccination.
- $+\nu S$: Those same individuals entering the V compartment.

3.2 Equilibrium Analysis (S^*)

At the disease-free equilibrium (where $I = 0$), the Susceptible population stabilizes at a level determined by the "tug-of-war" between births (μ) and vaccination (ν).

$$S^* = \frac{\mu N}{\mu + \nu}$$

This tells us that increasing the vaccination rate (ν) directly suppresses the equilibrium size of the susceptible pool.

4. Stochastic Control (Branching Processes)

In small populations or early outbreaks, we cannot rely on average rates. We deal with probabilities.

4.1 The Effect of Vaccination on Extinction

In a deterministic model, if $R_e = 1.1$, an epidemic is guaranteed. In a stochastic model, if $R_e = 1.1$, there is still a chance the outbreak dies out by luck.

Vaccination amplifies this "luck." Even if coverage is too low to achieve herd immunity ($R_e > 1$), it reduces the "force" of the outbreak (λ), increasing the probability of extinction (q).

4.2 Calculating Extinction Probability (q)

We assume the number of secondary cases follows a Poisson distribution with mean $\lambda = R_e$. The probability q that the chain of transmission breaks is the solution to:

$$q = e^{-\lambda(1-q)}$$

- If $\lambda \leq 1$: $q = 1$ (Extinction is certain).
- If $\lambda > 1$: $q < 1$ (There is a non-zero risk of a major epidemic).

Note: This equation cannot be solved algebraically. In the lab, you will use a numerical solver (like `uniroot` in R or `fsolve` in Python) to find q.

5. Strategic Comparisons

Strategy	Description	Pros	Cons
Routine / Prophylactic	Vaccinate continuous cohorts (e.g., infants) or mass campaign before outbreak.	Prevents epidemic start; protects most vulnerable.	Requires infrastructure; expensive; supply issues.
Reactive / Ring	Vaccinate contacts of identified cases.	Efficient use of doses; targets risk.	Requires excellent surveillance/contact tracing; often too slow.
Targeted	Vaccinate "Hubs" or "Super-spreaders" (high degree nodes).	Mathematically most efficient reduction in R_0.	Hard to identify who the "hubs" are; ethical concerns.

NPIs (Masks/Distance)	Reduce contact rate (β) rather than susceptibility (S).	Effective immediately; no medical supply chain needed.	High economic/social cost; compliance fatigue.

6. Symbol Reference Guide

Symbol	Pronunciation	Meaning	Unit/Range
S, I, R, V	Ess, Eye, Are, Vee	Compartments: Susceptible, Infectious, Recovered, Vaccinated	Count or Proportion
N	En	Total Population Size	Count
β	Bay-tuh	Transmission Coefficient	$Time^{-1}$
γ	Gam-uh	Recovery Rate (1/duration)	$Time^{-1}$
μ	Mew	Birth/Death Rate (demographics)	$Time^{-1}$
ν	New	Vaccination Rate (routine)	$Time^{-1}$
R_0	Are-nought	Basic Reproduction Number	Dimensionless
R_e	Are-ee	Effective Reproduction Number	Dimensionless
p	Pee	Vaccination Coverage	$0 \le p \le 1$
e	Ee	Vaccine Efficacy	$0 \le e \le 1$
p_c	Pee-see	Critical Vaccination Threshold	$0 \le p_c \le 1$
λ	Lam-duh	Mean Offspring (Poisson parameter)	Dimensionless
q	Cue	Probability of Extinction	$0 \le q \le 1$

7. Self-Assessment Questions

Conceptual

1. **Explain the difference between R_0 and R_e in the context of a vaccination campaign.**
   - *Answer:* R_0 is the potential spread in a totally susceptible population. R_e is the actual spread considering that some people are vaccinated/immune. Vaccination lowers R_e but does not change the biological R_0.
2. **Why might a targeted vaccination strategy (vaccinating high-contact people) require fewer doses than a random strategy?**
   - *Answer:* High-contact individuals contribute disproportionately to transmission. Removing them from the network breaks more potential transmission chains per dose than removing a

random, low-contact individual.

Calculation

3. Measles has an $R_0 \approx 15$. The available vaccine is 95% effective ($e = 0.95$). Calculate the critical coverage p_c.
   - *Step 1:* Calculate threshold for perfect vax: $1 - 1/15 = 0.933$.
   - *Step 2:* Adjust for efficacy: $0.933/0.95 = 0.982$.
   - *Result:* You need **98.2%** coverage.
4. A new flu strain has $R_0 = 1.5$. We have no vaccine, but NPIs (masks) reduce transmission (β) by 20%. Is this enough to stop the epidemic?
   - *Step 1:* New $R_{control} = R_0 \times (1 - 0.20)$.